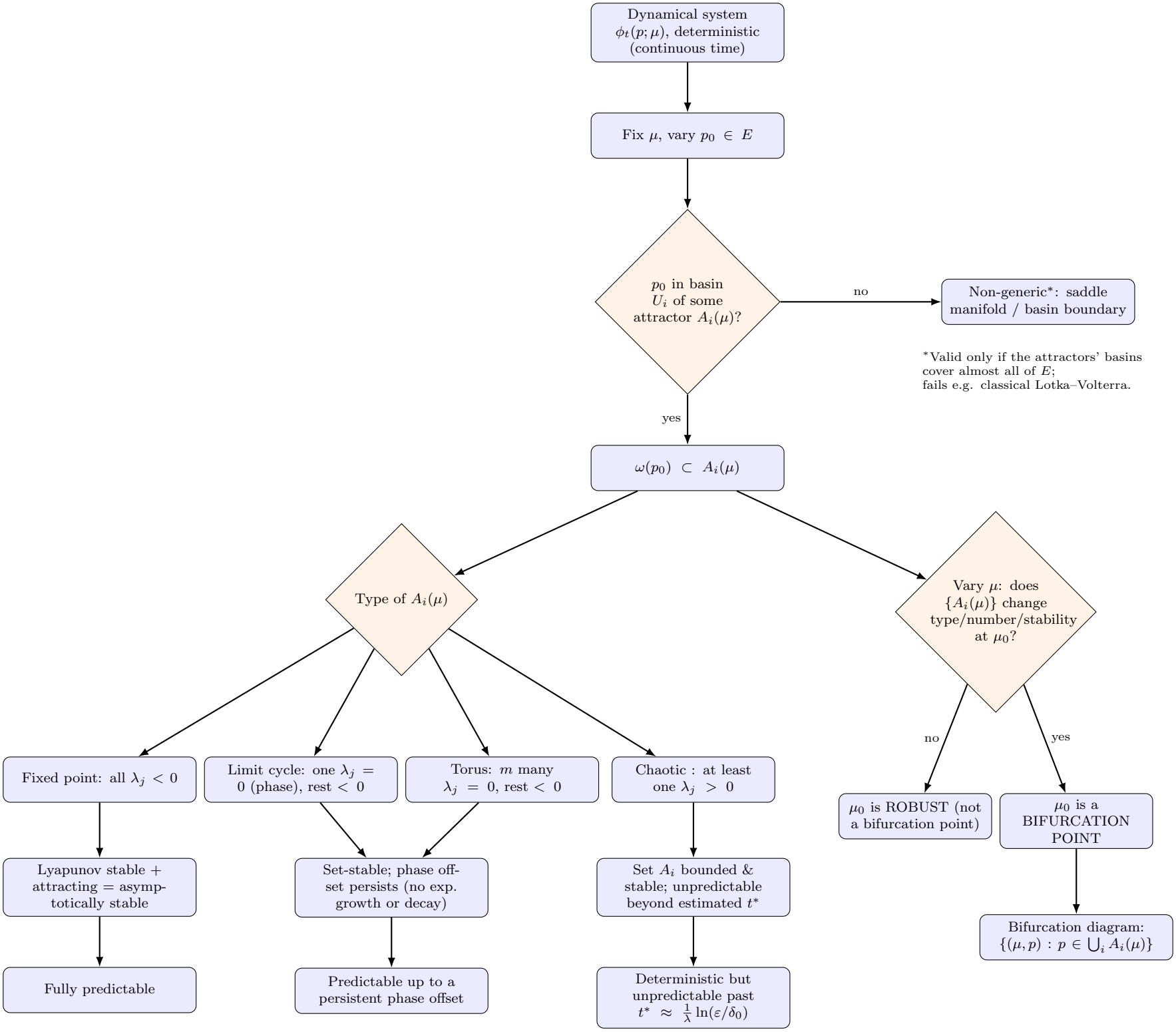

Dynamical system $\phi_t(p;\mu)$, deterministic (continuous time)
Fix μ, vary $p_0 \in E$
p_0 in basin U_i of some attractor $A_i(\mu)$?
no
Non-generic*: saddle manifold / basin boundary
*Valid only if the attractors' basins cover almost all of E; fails e.g. classical Lotka–Volterra.
yes
$\omega(p_0) \subset A_i(\mu)$
Type of $A_i(\mu)$
Vary μ: does $\{A_i(\mu)\}$ change type/number/stability at μ_0?
Fixed point: all $\lambda_j < 0$
Limit cycle: one $\lambda_j = 0$ (phase), rest < 0
Torus: m many $\lambda_j = 0$, rest < 0
Chaotic : at least one $\lambda_j > 0$
no
yes
μ_0 is ROBUST (not a bifurcation point)
μ_0 is a BIFURCATION POINT
Lyapunov stable + attracting = asymptotically stable
Set-stable; phase offset persists (no exp. growth or decay)
Set A_i bounded & stable; unpredictable beyond estimated t^*
Bifurcation diagram: $\{(\mu,p) : p \in \bigcup_i A_i(\mu)\}$
Fully predictable
Predictable up to a persistent phase offset
Deterministic but unpredictable past $t^* \approx \frac{1}{\lambda}\ln(\varepsilon/\delta_0)$